

Personal Lang Coach

System Architecture White Paper | Livo AI SWE Assessment | July 2026

1. Product And Architecture

Personal Lang Coach is a Vercel-hosted static web app for 15-45 second English speech uploads. The shipped design intentionally keeps audio processing inside the browser: Vercel serves HTML, CSS, and JavaScript, while the Web Audio API decodes the recording and the local analyzer produces scoring, mistake ranges, and coaching notes.

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User browser
-> file duration gate + explicit consent
-> Web Audio decoder
-> analyzer.js: voicing, pause, clarity, rhythm, pace
-> results renderer: score, waveform, segment/word highlights
Vercel static hosting serves assets only; no audio API or object store is used by default.
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2. Models, APIs, Scoring, And Highlights

Models and APIs: The implementation uses the browser Web Audio API and deterministic acoustic heuristics for the core score. Optional coaching can use Qwen3 4B or Gemma 3 4B through a Vercel API route to an OpenAI-compatible inference endpoint. The route sends the transcript and derived scores/issues, never raw audio. This preserves a no-key browser experience for baseline feedback while allowing richer coaching when a server-side model endpoint is configured.

Scoring: Audio is split into 30 ms frames with 15 ms hops. RMS energy estimates voicing and clarity; zero-crossing rate flags noisy consonant-like spans; pause gaps flag rhythm breaks. The overall score weights clarity 48%, rhythm 27%, pace 20%, and duration compliance 5%.

Highlighting: The app returns time-coded issues such as unclear segment, noisy articulation, uneven stress, long pause, or low speech density. If the learner supplies the expected transcript, issue time ranges are mapped onto words for word-level highlights; otherwise the same findings appear as segment-level coaching cards.

3. DPDP Compliance Posture

Voice recordings can identify a speaker and are treated as digital personal data. The app therefore uses data minimization by default: it does not upload, store, or log the audio. Processing happens only after an affirmative consent checkbox, and reset/page refresh clears the local object URL, decoded buffer, transcript, and analysis state. Optional model coaching requires separate consent before sending transcript and derived assessment data.

Storage and retention: No server-side audio storage exists in the default deployment. Retention is limited to browser memory for the active session; the specified purpose ends when feedback is shown or the user resets.

Residency and processors: Because audio is not transmitted, there is no audio data residency dependency. Vercel only hosts static assets. If a production STT service is added, it should use a contracted processor, approved transfer/residency posture, encryption, short TTL storage, deletion APIs, and logs that exclude raw audio.

Rights and deletion: The UI exposes plain-language notice, purpose limitation, and deletion by reset. A production account-based version would add an authenticated deletion request flow, grievance contact, and audit trail.

4. Trade-offs And Next Week

The major trade-off is privacy and deployability over phoneme-level accuracy: the current app cannot prove a specific word was mispronounced without a transcript or STT alignment. It still provides useful learner feedback on fluency, pauses, clarity, pacing, and unstable stress. With another week, I would add optional server-side forced alignment, phoneme comparison against a target passage, calibrated CEFR-style scoring, signed transient uploads, and a teacher review dashboard.

Compliance reference: Digital Personal Data Protection Act, 2023, Gazette of India, Act No. 22 of 2023. Relevant controls: notice and consent, security safeguards, erasure, and cross-border transfer restrictions.